Convergence Tests

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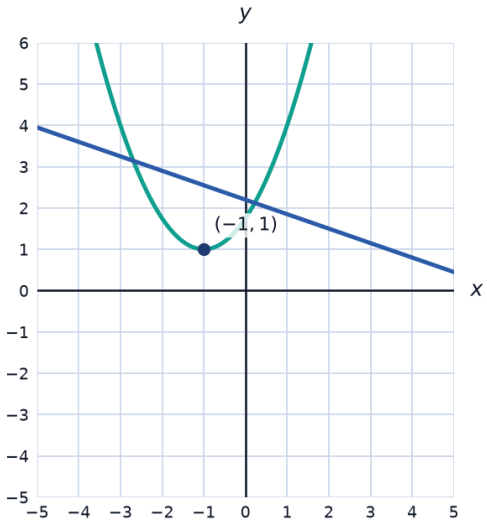
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The ratio and comparison tests**

Use the ratio test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.

2 **[Foundational]** • **Target: The ratio and comparison tests**

Use the ratio test on $\sum_{n=1}^{\infty} \frac{n!}{n^n}$.



3 **[Foundational]** • **Target: The ratio and comparison tests**

Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+3}$ converges, by comparing to $\sum \frac{1}{n^2}$.

4 **[Foundational]** • **Target: Alternating series and the p-series test**

Determine whether each p-series converges or diverges:

a $\sum_{n=1}^{\infty} \frac{1}{n^3}$

b $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

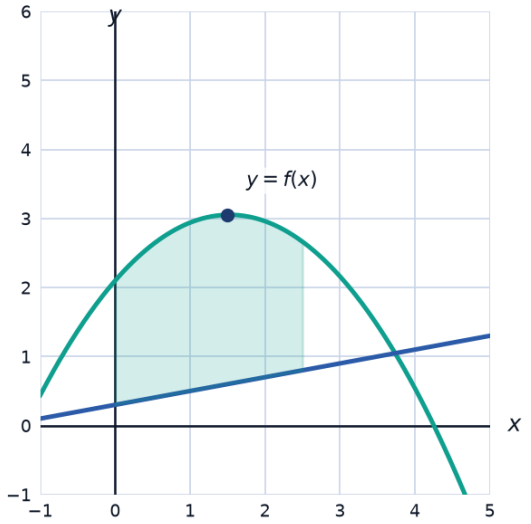
5 **[Foundational]** • **Target: Alternating series and the p-series test**

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges, and classify it as absolutely convergent, conditionally convergent, or divergent.

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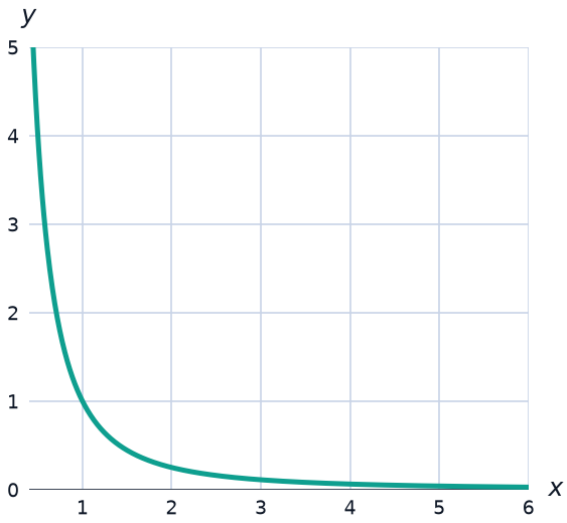
6 [Foundational] • Target: The integral test

Use the integral test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n \ln n}$ (starting at $n = 2$) converges or diverges, using $\int_{2}^{\infty} \frac{1}{x \ln x}\,dx$.



7 [Foundational] • Target: The integral test

Use the integral test to confirm that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, using $\int_{1}^{\infty} \frac{1}{x^2}\,dx$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Choosing the right test

For each series, name the most efficient test to apply (ratio, comparison, p-series, or alternating series), without fully evaluating:

a $\sum \frac{2^n}{n!}$

b $\sum \frac{(-1)^n}{\sqrt{n}}$

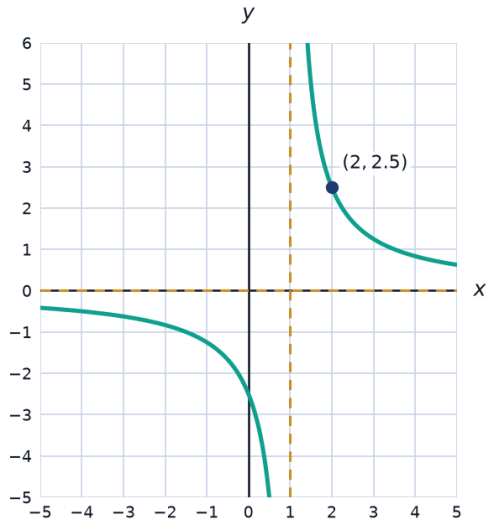
c $\sum \frac{1}{n^4}$

9 [Intermediate] • Target: The limit comparison test

Use the limit comparison test to determine whether $\sum_{n=1}^{\infty} \frac{n+1}{n^3+2}$ converges, comparing to $\sum \frac{1}{n^2}$.

10 [Intermediate] • Target: The limit comparison test

Use the limit comparison test on $\sum_{n=1}^{\infty} \frac{3n^2-1}{n^4+2n}$, comparing to $\sum \frac{1}{n^2}$.



11 [Intermediate] • Target: The root test

Use the root test to determine whether $\sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$ converges.

12 [Intermediate] • Target: The root test

Use the root test on $\sum_{n=1}^{\infty} \frac{3^n}{n^n}$.

13 [Intermediate] • Target: Absolute versus conditional convergence

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ is absolutely convergent, conditionally convergent, or divergent.

14 [Intermediate] • Target: Estimating the sum of an alternating series

The alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ converges by the alternating series test. Use the fact that the error after n terms is bounded by the first omitted term to estimate how many terms are needed for the partial sum to be within 0.001 of the true sum.

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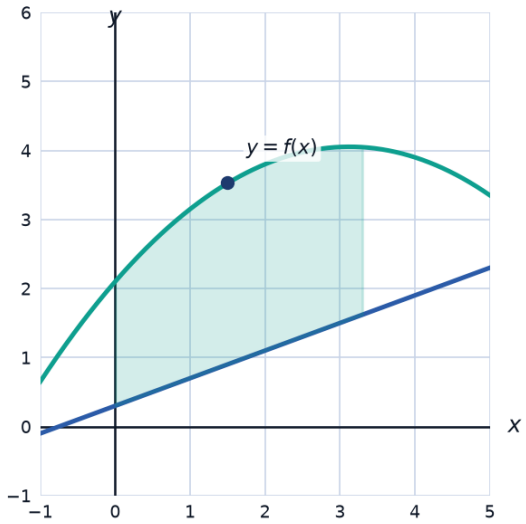
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: A combined convergence-test challenge • Cross-Lesson: 7.01 Integration by Substitution**

Determine whether $\sum_{n=1}^{\infty} \frac{n^2}{3^n}$ converges, using the ratio test.

16 **[Challenging] • Target: A combined convergence-test challenge • Cross-Lesson: 7.02 Integration by Parts**

Determine whether $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges, using the integral test with $u = \ln x$.



17 **[Challenging] • Target: A combined convergence-test challenge • Cross-Lesson: 8.01 Area Between Curves**

A student claims that since $\lim_{n\to\infty} \frac{1}{n} = 0$, the series $\sum_{n=1}^{\infty} \frac{1}{n}$ must converge. Explain what is wrong with this reasoning, and state the correct necessary (but not sufficient) condition that the nth-term test actually provides, using the harmonic series as a counterexample.

18 **[Challenging] • Target: A multi-step convergence investigation • Cross-Lesson: 8.02 Volumes of Revolution**

For the series $\sum_{n=1}^{\infty} \frac{n^3 + 2n}{2n^5 - n + 1}$, first find the nth-term limit to check the necessary condition, then use the limit comparison test with $\sum \frac{1}{n^2}$ to determine whether the series converges.

19 **[Challenging] • Target: A multi-step convergence investigation • Cross-Lesson: 8.03 Arc Length and Improper Integrals**

A quality-control engineer models cumulative defect probability using $\sum_{n=1}^{\infty} \frac{1}{n \cdot 2^n}$. Use the ratio test to confirm this series converges, and briefly explain (in one sentence) why a convergent bound like this is useful for guaranteeing that a manufacturing process's total defect risk stays finite over infinitely many production runs.

20 **[Challenging] • Target: A multi-step convergence investigation • Cross-Lesson: 9.01 Separable Differential Equations**

Level 1 · Foundations Use the Divergence Test to determine whether $\sum_{n=1}^{\infty} \cos\left(\frac{1}{n}\right)$ converges or diverges.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The ratio and comparison tests

Series $\sum_{n=1}^{\infty} (-1)^{n-1}/\sqrt{n}$ converges by alternating series test. Is it absolutely convergent?

a. Yes, $p = 1/2 < 1$ diverges

b. No, $\sum 1/\sqrt{n}$ diverges ($p = 1/2 \leq 1$)

c. Yes, $\sum 1/\sqrt{n}$ converges

d. Cannot determine

22 [Foundational] • Target: Alternating series and the p-series test

Integral test convergence of $\sum_{n=1}^{\infty} 1/n(\ln n)^2$ (from $n = 2$) gives sum approximately:

a. converges

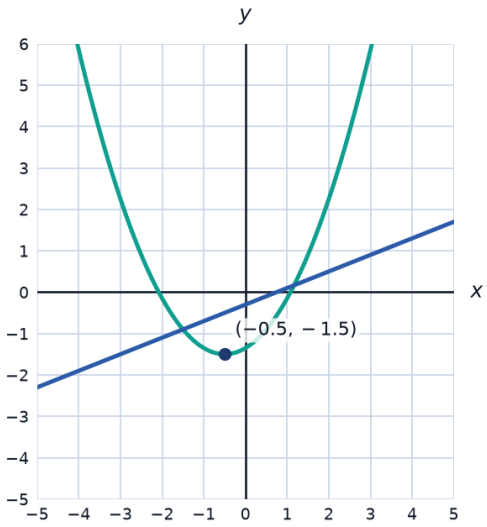
b. diverges

c. equals e

d. equals π^2

23 [Foundational] • Target: The integral test

Which test determines $\sum n^2/e^n$? Answer: by ratio test $L =$



a. e

b. 1

c. $1/e < 1$

d. e^2

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24 [Foundational] • Target: Choosing the right test

Series $\sum_{n=1}^{\infty} n \cdot (1/2)^n$: ratio test gives $L =$

a. 1

b. 2

c. 1/2

d. $n/2$

25 [Foundational] • Target: The limit comparison test

Use the integral test to show $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges. The integral $\int_2^{\infty} \frac{dx}{x(\ln x)^2} =$

a. ∞

b. π

c. $\ln 2$

d. $1/\ln 2$

26 [Foundational] • Target: The root test

The ratio test gives $L = 1$: the test is:

a. inconclusive

b. divergent

c. convergent

d. absolutely convergent

27 [Foundational] • Target: Absolute versus conditional convergence

Limit comparison: $a_n = 1/(n^2 + 1)$, compare to $b_n = 1/n^2$. $\lim a_n/b_n =$

a. 0

b. 1/2

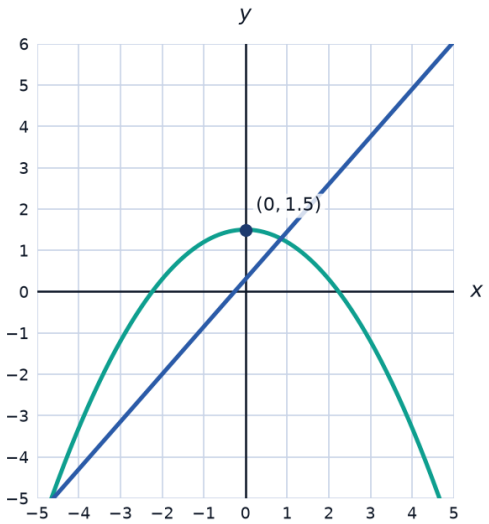
c. ∞

d. 1

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28 [Foundational] • Target: Estimating the sum of an alternating series

Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{(2x)^n}{n!}$.



a. 0

b. 2

c. ∞

d. $\frac{1}{2}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A combined convergence-test challenge

Convergence of $\sum n^2/3^n$: by ratio test, $L =$

a. 1

b. 1/3

c. 3

d. 0

30 [Intermediate] • Target: A multi-step convergence investigation

Alternating test: $a_n = 1/n!$: does $a_n \to 0$? Does $a_{n+1} < a_n$?

a. Neither

b. Both $a_n \to 0$ and $a_{n+1} < a_n$

c. $a_n \to 0$ only

d. $a_{n+1} > a_n$

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31 [Intermediate] • Target: The ratio and comparison tests

Comparison: if $\sum b_n$ diverges and $a_n \geq b_n > 0$, then $\sum a_n$:

a. converges

b. equals $\sum b_n$

c. is inconclusive

d. diverges

32 [Intermediate] • Target: Alternating series and the p-series test

The integral $\int_0^1 e^{-x^2} dx$ (error function) — the exact value cannot be expressed in elementary functions. Using the series, approximate as $\sum \frac{(-1)^n}{n!\,(2n+1)}$. First two terms:

a. $1 - 1/6 \approx 0.833$

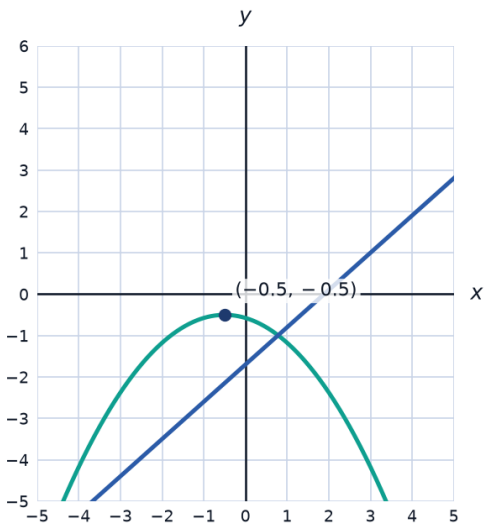
b. $1 - 1/3 \approx 0.667$

c. $1 + 1/3 \approx 1.333$

d. 0

33 [Intermediate] • Target: The integral test

Comparison: $0 \leq a_n \leq 1/n^2$. Since $\sum 1/n^2$ converges, $\sum a_n$:



a. diverges

b. may diverge

c. converges

d. oscillates

34 [Intermediate] • Target: Choosing the right test

Bessel function $J_0(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2}\left(\frac{x}{2}\right)^{2m}$. Radius of convergence:

a. 1

b. ∞

c. π

d. 2

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35 [Intermediate] • Target: The limit comparison test

Absolute convergence implies conditional convergence. The converse:

a. is also true

b. is false (counterexample: alternating harmonic series)

c. requires $a_n \geq 0$

d. is true for power series

36 [Intermediate] • Target: The root test

Consider the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$. If the radius of convergence is $R = 3$ and the center of the series is $a = 2$, which of the following intervals represents the values of x for which the series converges?



a. $(-1, 5)$

b. $[-1, 5]$

c. $(-1, 5]$

d. $[-1, 5)$

37 [Intermediate] • Target: Absolute versus conditional convergence

Apply the ratio test to $\sum_{n=1}^{\infty} \frac{6^n}{n!}$. What is $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$?

a. 0

b. ∞

c. 1

d. 6

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Estimating the sum of an alternating series • Cross-Lesson: 8.02 Volumes of Revolution

Consider the power series $\sum_{n=1}^{\infty} \frac{(2x-1)^n}{n \cdot 3^n}$. For what values of x does this series converge conditionally?

a. $-1 \leq x < 2$

b. $-1 < x \leq 2$

c. $-1 \leq x \leq 2$

d. $-1 < x < 2$

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39 [Challenging] • Target: A combined convergence-test challenge • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Consider the power series $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n^2 3^n}$. This series defines a function $f(x)$ over its interval of convergence. Let I be the interval of convergence for this power series. Determine which of the following statements is TRUE about $f(x)$ and its behavior at the endpoints of I.

a. The radius of convergence is 3, and the series converges at both endpoints of the interval of convergence.

b. The radius of convergence is 3, and the series diverges at the left endpoint and converges at the right endpoint.

c. The radius of convergence is $1/3$, and the series converges at both endpoints of the interval of convergence.

d. The radius of convergence is $1/3$, and the series diverges at both endpoints of the interval of convergence.

40 [Challenging] • Target: A multi-step convergence investigation • Cross-Lesson: 9.01 Separable Differential Equations

Find the interval of convergence of $\sum \frac{(x-1)^n}{n3^n}$:

a. $[-2, 4)$

b. $(-2, 4)$

c. $(-2, 4]$

d. $[-2, 4]$

41 [Challenging] • Target: The ratio and comparison tests • Cross-Lesson: 1.01 Precalculus Review for Calculus

The error function $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$. Using $e^{-t^2} = \sum (-1)^n t^{2n}/n!$:

a. $\operatorname{erf}(x) = \sum x^n/n!$

b. $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{n!\,(2n+1)}$

c. $\operatorname{erf}(x) = e^{-x^2}$

d. $\operatorname{erf}(x) = 2x/\sqrt{\pi}$

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42 [Challenging] • Target: Alternating series and the p-series test • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

For a power series $\sum c_n x^n$, the radius of convergence R satisfies $1/R =$

a. $\limsup |c_n|^{1/n}$

b. $\limsup |c_{n+1}/c_n|$

c. $\sup |c_n|$

d. $|c_0|$

43 [Challenging] • Target: The integral test • Cross-Lesson: 11.01 Sequences and Series

Integral test: $\sum_{n=1}^{\infty} f(n)$ converges iff $\int_1^{\infty} f(x)\,dx$ converges, when f is:

a. any function

b. bounded

c. increasing

d. positive, decreasing, continuous

44 [Challenging] • Target: Choosing the right test • Cross-Lesson: 12.02 Power Series and Taylor Series

Using the power series $1/(1-x) = \sum x^n$, find $\sum n x^{n-1} =$

a. $1/(1-x)$

b. $1/(1-x)^2$

c. $x/(1-x)^2$

d. $1/(1+x)^2$

45 [Challenging] • Target: The limit comparison test • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

The series $\sum_{n=0}^{\infty} r^n$ converges for $|r| < 1$ to:

a. $1/(1-r)$

b. $r/(1-r)$

c. $1/(1+r)$

d. r

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